AGRO-BHARAT: DEVELOPMENT OF A PRODUCT WITH FERTILIZER FUNCTION FORMULATED FROM INDIAN NEEM PROPERTIES

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Abstract

Nature-Based Solutions (NBS) focus on solving environmental problems. Among these problems, the search for alternative fertilizers that do not pollute the environment stands out. This study sought to select and apply natural compounds that could reduce or replace the use of commercially available agribusiness products. The plant commonly known as Indian Neem (*Azadirachta indica*) has been shown to be a good potential fertilizer candidate. Thus, the objective of this study was to develop a new product capable of providing the nutrients necessary for bell pepper growth and development, as well as acting as an agricultural pesticide. To this end, the methodology was divided into 5 stages: extraction of the respective Neem compounds; preparation of the formulations; analysis of the formulation; application of the product to bell pepper seedlings and collection and statistical analysis of the data. The active ingredients of neem were extracted and applied to the plants. Analysis of the data has shown that the growth of the group treated with Neem was significantly more robust than that of the control group. The plants treated with Neem have grown faster than those in the control group. Thin-layer chromatography (TLC), performed in a partner laboratory, showed that the neem extract has a greater number of compounds and is therefore more complex than common fertilizers. Therefore, it can be concluded that this study could contribute to the development of a new, more sustainable product.

Keywords: Fertilizers; Crop Protection Products; Indian Neem; Bell Pepper; Agricultural Engineering.

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1. INTRODUCTION

Brazil has plentiful potential for agribusiness due to its extensive territorial area, soil quality, climatic conditions, among other reasons. Commodities generate significant revenues due to their exports (1). The planting of peppers, as well as other crops, involves the use of chemical compounds with potential risks to the environment.

1.1 Fertilizers

Fertilizers provide key nutrients and promote plant growth. They contain both macronutrients and micronutrients essential for plant development.

Commonly marketed fertilizers, such as NPK, supply crucial macronutrients (N, P, K, Ca, Mg, and S). Nitrogen, for instance, is not directly assimilated by plants. While the nitrogen cycle naturally synthesizes amine and amide groups through ammonia, the Haber-Bosch process (2) enables artificial nitrogen production, considerably boosting agricultural productivity. However, this improvement has also introduced new risks, such as leaching, CO_2 emissions, and ammonia volatilization in the soil.

Micronutrients such as B, Zn, Cu, Fe, Mo, Cl, and Mn influence plant growth. In this sense, the yield of organic cultivation - that is, without the addition of industrial compounds - can achieve the same performance of plantations in which NPK is already applied (3).

In addition, the recent Micro Carbon Technology MCT® methodology (4), which can boost the encapsulation of organic matter, has the potential to revolutionize agriculture. By employing reticles made of polymers and colloids, encapsulated nutrients can remain in the medium and mitigate photodegradation by heat, ion oxidation, and UV rays (5).

Nanofertilizers contain materials rich in humic substances added to the soil, which can infiltrate the earth, supplementing the metabolism of plants (6). In conclusion, nature-based solutions are relevant in Agricultural Sciences because of their economic and environmental implications.

1.2 Nature-based solutions and alternative agricultural methods

All the factors already mentioned, among others, have led to the growing consumption of organic products and the investment in producing new biotechnologies to support agricultural development (7). An example of biotechnological practice is the use of genetically modified seeds that can naturally resist attacks because they have in their composition the addition of a substance considered toxic to predators.

Other alternatives that may be helpful in agricultural production are so-called nature-based solutions, also known as NBS. This term has been used internationally and denotes the conservation, restoration, and management of ecosystems sustainably. It encompasses ways of understanding issues related to the ecological protection of the natural environment (8).

NBS can encompass various environmental initiatives and forms of ecosystem-based climate change adaptation, ecological restoration, and protection of conservation areas. These actions can occur in different environments and in diverse ways (8). In this context, medicinal plants, their seeds, leaves, etc., can function as raw materials for active ingredient extraction. Also, it is an alternative to circumvent the contamination problem related to synthetic compounds used as pesticides and fertilizers.

Some plant types may contain active ingredients capable of performing the function of a fertilizer, as is the case of the Indian Neem, or even have substances capable of acting as pesticides, as some plant oils (3). The following section describes Neem characteristics.

1.3 Indian Neem

Indian Neem is an Asian plant known and utilized in India for over two thousand years. It has a wide range of applications, including fertilizer, medicine, and in the production of cosmetics, among many others. Currently, this plant's cultivation - also known as Amargosa - is central in countries of Central America, Africa, the United States, and Australia (9). Figure 01 shows the plant's appearance.



Figure 01 - Example of the Indian Neem plant, with its evident fruits and leaves.

Neem's scientific name is *Azadirachta indica*. It is a large tree of the order Sapindales and family Meliaceae (9). By isolating and obtaining a derivative of the tree, Azadirachtin, it is possible to explore its unique properties. This compound is non-flammable, non-corrosive, and formulated as an emulsifiable concentrate. Additionally, it acts systemically without posing significant environmental risks.

Azadirachtin is the main compound found in *Azadirachta indica* extracts. However, azadirachtin goes through the bio-elimination process, degrading when exposed to light. After 24 hours, although there are still remnants of the compound, they solely affect invertebrates. Thus, over time, before being consumed by humans, azadirachtin undergoes photodegradation and is diluted in water, being found at a concentration of 12 µg/mL, lower than the maximum concentration recommended by the Brazilian Health Regulatory Agency (ANVISA) 2016 technical standard. The ingested dose only becomes harmful above 1600 mg/kg/day (10). Neem was introduced in Brazil approximately 30 years ago. According to LIMA (11), Neem's uses range from oils and products derived from its leaves to the tree's bark.

Planting Neem requires careful steps and precautions to ensure the supply of nutrients and satisfactory ecological maintenance. (9) (12)

Furthermore, the plant also has medicinal effects, presenting hypolipidemic, antifertility, microbicides, antidiabetic, hepatoprotective, antipyretic, insecticide, nematicides, anti-ulcers, antioxidants, neuroprotective, cardioprotective, and antileishmanial functions (13).

Significantly, with the plurality of possible effects of Neem, the plant undergoes purity tests to ensure that its dosage provides the maximum benefits. In addition to not posing a health threat, the product presents low toxicity risk and can be inhaled by humans. In addition, neem is safe in contact with the skin since it has dermatological properties. It can also be ingested in the form of herbal teas and is present in the composition of common hygiene products (14). Other compounds present in the plant are Nimbin, Nimbidin, Salanine, and Quercetin. Accordingly, these components are associated with repellent properties and may be involved in combating diseases and infestations.

Figure 02 shows two examples of molecules in Neem, Salanine, and Nimbin, present mainly in its leaves (15). In addition, the presence of, among other particles, organic contents that affect the properties of Neem stands out: carboxylic acids such as glutamic acid, aspartic acid, margaric acid, and several other long-chain fatty acids. Neem also contains the amino

acids alanine and tyrosine, which contribute to the functional structuring that enables their oxidation-reduction with their oxygen atoms in the middle of their chains - heteroatoms - as indicated by Figure 02 (16).

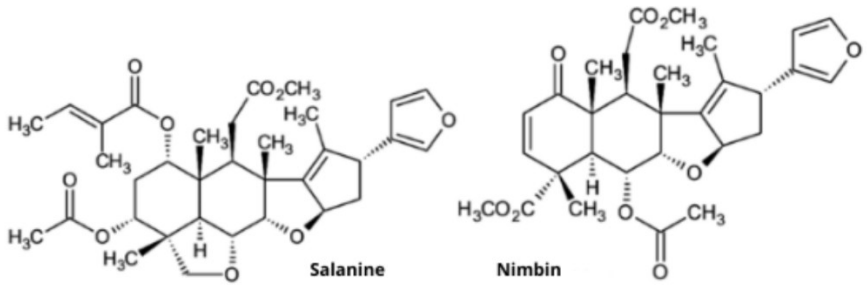


Figure 02 - Chemical formula of the main active principles of Indian neem, being represented by limonoids.

In addition to the compounds mentioned above, there are limonoids - limonene derivatives characterized by the presence of several organic aromatic rings - and other triterpenes in Neem, which can act as an insecticide (17). This action is also due to the presence of Nimbin, a limonoid that potentializes adverse effects on pests - influencing their growth and reproduction and causing muscle weakening and dysphagia (18), thereby increasing agricultural productivity.

Neem also has secondary metabolites that can be identified using high-performance liquid chromatography (HPLC) and gas chromatography (GC) to locate the properties of its extracts and subsequently isolate and concentrate compounds. Mass spectrometry and nuclear magnetic resonance (NMR) are part of the analysis to identify the active components (19).

These compounds have the potential to function as beneficial health products, acting as antioxidants through their high total content of phenolic compounds and flavonoids - which are polyphenolic compounds with two aromatic rings linked to three carbons in a chain. Figure 03 represents the general structure of flavonoids that make up the plant, in which the organic formula could vary depending on the amounts of phytopharmaceuticals involved

and the polyphenolic content (18).

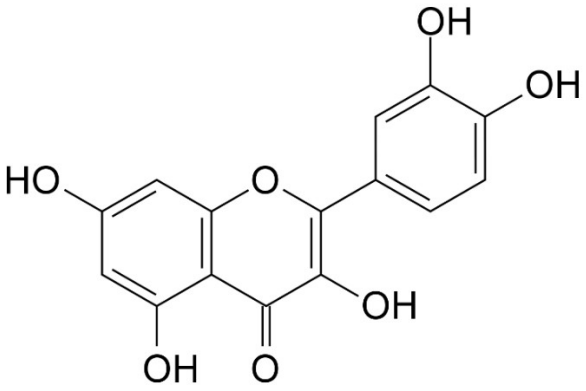


Figure 03 - Organic formula of a flavonoid molecule.

Complementing the aforementioned point, studies in academic literature feature the diversity of compounds in the different derivatives of the neem plant: oil, bark, roots, seeds, leaves, fruits, and flowers. Figure 04 was adapted from BATRA (20). These plant units reveal unique nutritional aspects depending on each branch of the Neem tree. On another level, the species has variable nutritional content across its various plant types, showing the quality of seedlings produced with macronutrient treatment (N, P, K, Ca, Mg, and S) and micronutrients (B, Cu, Fe, Mn, and Zn). (21)

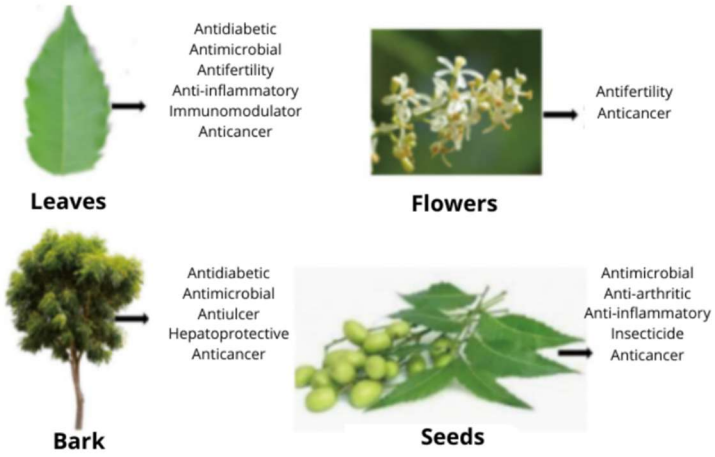


Figure 04 - Organization and systematization of functions of each part of the Neem tree. Referring, respectively, to the purpose of the leaf extract; of the properties of the flowers, of the trunk and its bark, and of the fruit and its seeds.

Neem leaves also contain several organic compounds. Neem retains key nutrients such as nitrogen, phosphorus, potassium, calcium, and magnesium (22), which can act as natural fertilizers. In addition to proving the benefits of using oil from its seed, extracted by cold pressing, the use of neem and its derivatives as fertilizer remains poorly explored. After obtaining the oil from the kernel of the seed, the seed mash – crushed material from the seed shell – can be used as an organic fertilizer. Therefore, its fertilizing potential, even though it is less potent compared to Aloe Vera, for example, is still quite significant, as the leaves of the plant have a high protein content (7.1%), carbohydrates (22.9%), minerals, calcium, phosphorus, vitamin C, carotene and amino acids, such as glutamic acid, tyrosine, alanine, aspartic acid, glutamine, cysteine and also fatty acids, which makes it suitable for the performance of purification processes that isolate active ingredients (17).

Compared to the NPK-based compounds seen previously, the structure of Neem can give rise to nano fertilizers and nano pesticides that are alternative solutions that sustainably enhance its properties in agriculture, which can prevent the leaching of excess nutrients from synthetic compounds and reduce the effects of eutrophication on plant cultures (23). These characteristics indicate the possibility of using Neem in the nascent technology of nanoparticles to perform an agricultural treatment on a microscopic scale. By improving over time, encapsulation techniques have achieved greater efficiency (3).

1.4 Bell pepper production in Brazil

Bell pepper is a species of vegetable crop that has significant value in the cuisine of tropical countries. The development of the plant to its average fruiting state encounters obstacles in nutritional requirements, sensitivity, and agricultural care (24). Its consumption is usually ‘in natura’, which subjects the fragile crop to agrarian losses and the difficulty of following standards related to the exposure to pesticides tolerated by health inspection agencies (25). The main concentration of the crops is in the state of São Paulo, with a more significant number of cultivable hectares compared to the other Brazilian states.

The vegetable has a shorter ripening time compared to other herbaceous plants, providing a favorable financial return for commercialization in Brazil, notably green bell peppers. The color of this vegetable varies as it advances in its state of maturation; thus, it can be green, red, yellow, orange, cream, and purple (26).

Among the main species that attack pepper agriculture are aphids, mites, cows, and lizards. In this context, one of the pest species that most affects the crop is *Icerya purchasi*, known

as the white aphid. Furthermore, other variations of arthropod pests are considered threats, as they are prone to contaminating plants and their leaves by spreading diseases that reduce productivity (27).

Plants of the genus *Capsicum*, in particular, are known for the burning sensation they produce when consumed, as they contain Capsaicin. In addition to being natural sources of flavonoids (as discussed in Section 1.5), bell peppers can have carotenoids and are an essential source of vitamin C (28) but are not as spicy as peppers since they contain comparatively less Capsaicin.

The place where the bell pepper grows can affect its quality and the color of its leaves. The acidity of the culture substrates – or basicity – is a parameter that drives soil volatilization to affect the nutritional composition and the soil's pH (29). The evaluation of bell pepper's quality considers the dose of vitamins that remained in its culture media, in addition to the final biomass obtained, and residues that were absorbed from pesticides by the vegetable, as can be seen in Figure 05, stressing that pepper surpasses the 90% mark of the highest residual of pesticides that are grown in Brazil.

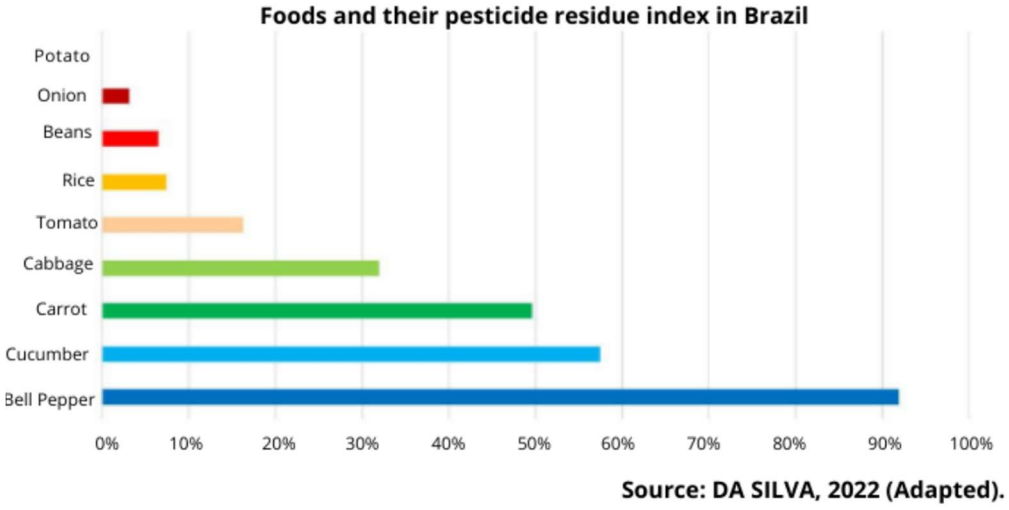


Figure 05 - Foods with the highest residual of pesticides that are grown in Brazil.

Relevant research shows that the pepper grown in Brazil has a concentration of 4.31 kg/ha of pesticides (30). The country has large arable areas compared to other countries that are agricultural exporters. Overall, this number is considered high. Climatic conditions such as humidity and temperature can significantly affect plant development, influencing pest activity to varying extents (31). In turn, these factors contribute to differences in nutritional

and mineral dosages in each region of the world and in Brazil, both due to the influence of luminosity, temperature, soil composition, and relative humidity, which can result in a higher planting yield.

2. INQUIRY QUESTION

How to develop a new product based on natural resources, which can be applied in the cultivation of peppers and that acts as fertilizer in natural agriculture?

3. HYPOTHESIS

According to LIMA (11), Neem, a plant native to Asia, has medicinal and herbal properties, insecticidal effects, and, above all, contains several nutrients, making it a potential fertilizer.

Thus, this work aims to develop a formulation of a product that can act as a fertilizer. This product will have Indian Neem as a raw material. We hypothesize that the nutrients present in the Neem extract may assist in the growth and development of a plant.

Bell peppers were chosen for cultivation in the present study because they are among the ten most important vegetables in terms of economic value in Brazil and are considered one of the products that consume the most pesticides in the country. (31)

4. OBJECTIVES

4.1 General Objective

The general objective of this study is to develop a new product derived from the Neem plant, capable of providing the necessary nutrients for bell pepper growth and development.

4.2 Specific Objectives

a) Obtain the Indian neem extract with ethanol and perform a composition analysis of the solution;

b) Sow a large sample of pepper seedlings in a controlled environment;

c) Apply the extracts in the cultivation of peppers;

d) Analyze the experimental groups of pepper cultivation according to the parameters stem growth and number of leaves, over a period of 10 weeks.

5. MATERIALS AND METHODS

5.1 Extraction of the active ingredients of Neem

The extraction of Indian Neem (whose active ingredient – mainly azadirachtin – is present in the *Azadirachta indica* plant) used an adapted methodology (32). Briefly, the soluble compounds from the leaves were obtained using ethyl alcohol, according to the steps detailed below. The location was the chemistry laboratory of *Colégio Dante Alighieri*. The neem leaves were weighed and dried until a uniform mass was obtained. The brand of neem obtained was *Grupo Rocha Saúde*® (Figure 06).



Figure 06 - Sample of commercial Indian neem used for the extraction of the active ingredients.

In the study, we measured the capsule's mass used in the drying stage. Subsequently, 50g of neem leaves were placed in an oven at 85ºC for 8 hours (Figure 07). This temperature was chosen (33) so that the azadirachtin half-life of 12 hours at 90°C does not affect its concentration. The moisture content present in the leaves was determined by the gravimetric method (measurement of the mass of the material before and after the drying stage). The drying stage was repeated three times until the test material felt dry to the touch.

The dried material was placed in a desiccator to prevent moisture reabsorption.



Figure 07 - Experimental steps related to the extraction of the main compounds present in Indian neem, carried out in the chemistry laboratory of Colégio Dante Alighieri.

A volume of 200 mL of PA ethyl alcohol *(Itajá® - 99ºGL)* was added to the dried material. No water was used in the process, as it reduces the solubility of the compounds of interest at the extraction time (32). In this process, 50g of dried leaves were blended with the ethyl alcohol solution for 4 minutes using a *Britania® Diamante* blender, model 800, 900W (Joinville, Brazil). After blending, the suspension underwent filtration, which, in turn, was divided into two stages: first, filtration using the sieve, and second, filtration using laboratory filter paper. The final stage consisted of drying the filtered material at 38ºC using a *FANEM®* oven, model 515 (Guarulhos, Brazil). This temperature prevents the degradation of the compounds. Finally, Figure 08 illustrates the experimental steps of the neem extraction method.



Figure 08 - Extraction steps of Indian Neem carried out in the chemistry laboratory of Colégio Dante Alighieri: a) Sieving of the extract + neem leaves; b) Filtration with filter paper; c)

Appearance of the extract obtained (dark green).



Figure 09 - Appearance of Indian Neem extract after ethanol

Figure 09 shows the appearance of the neem extract placed in a capsule and submitted to the greenhouse. The solid material obtained after drying had a mass of 0.1977 g (Figure 10).



Figure 10 - Weighing of neem extract. Chemistry laboratory of Colégio Dante Alighieri.

5.2 Preparation of formulations

The solid samples of neem (with a mass of 0.1977 g) were diluted in 50 mL of water. Subsequently, the solution containing neem extract had a final volume of 200 mL (after adding water). Solution 2, containing just water, was given to the control group. Table 01 shows a summary of these formulations.

Table 01 - Formulations used in this study from the extracts of Neem and Control.

Solution Number	Formulation composition
1	Neem Extract with Water
2	Water

Source: Authorship.

5.3 Application of products obtained on bell pepper

Initially, the bell pepper (*Capsicum Annuum*) seeds were planted in the biotechnology lab of Colegio Dante Alighieri school and separated into 2 experimental groups – the test and control groups (Figure 11).



Figure 11 - Preparation of the pepper pots at the Laboratory of Biotechnology of the Dante Alighieri School.

The solution application on each bell pepper seedling started in March and extended throughout April and June, adding up to a total of 10 weeks. The next step consisted of performing the following procedures: watering each seedling daily and conducting the application of the two experimental groups with aliquots of 1 mL of the respective products directly on the root of each seedling. Each experimental group contained 2 (two) bell pepper seedlings in its pot. It is worth noting that the seedlings were not watered on Sundays and during a holiday, as the school was closed on those days and the lab is a locked and secure room.

6. RESULTS AND DISCUSSION

6.1 Obtaining the Neem extract

The study's first stage, as mentioned in a previous section, consisted of drying Neem leaves until a uniform mass precipitate was obtained. The results of this step are in Table 02.

Table 02 - Results of the Neem drying stage performed in the chemistry laboratory of Colégio Dante Alighieri.

Drying stage	Capsule Mass	Weight before drying	Weight after drying
1	302.46 g	50.0 g	42.59 g
2	302.46 g	42.59 g	42.57 g
3	302.46 g	42.57 g	42.36 g

Source: Authorship.

The mass loss of neem leaves can be between 20% to 75% (34). In this study, the total mass loss was 15.28%, lower than expected.

The Neem solution contains a volume of 190 mL, which will be applied later in the bell peppers. The next session will detail the experimental groups.

6.2 Planning of experimental groups

The names of the experimental groups assembled, the composition of the test and control dosages applied, and their identification for this study are summarized in Table 03.

Table 03 - Experimental groups planned for the experiment to evaluate Neem and Control extracts in pepper seedlings.

Experimental Groups	Composition	Acronym
Control Group	Water	COG
Nim Group	Neem dissolved in water	NGR

Source: Authorship.

The data-gathering process occurred weekly, according to pre-established parameter plant height.

6.3 Analysis of the formulation

6.3.1 Thin layer chromatography

The Neem extract obtained in the laboratory was evaluated in a partner laboratory, with the assistance of researchers in the field of natural products, Dr. Anelize Bauermeister and Dr. Massuo Jorge Kato, from the Institute of Chemistry at the University of São Paulo (IQ-USP). For this evaluation, methanol was used as a solvent in the experimental procedures, instead of ethyl alcohol. Since methanol has a high toxicity, it cannot be used in a high-school setting. Methanol extracts present some differences compared to ethanol. Methanol evaporates faster and is slightly more polar than ethanol. Methanol may be better for the extraction of highly polar compounds (such as flavonoids, phenolics, and glycosides). Ethanol can dissolve moderately polar to nonpolar constituents (such as terpenoids, alkaloids, and aglycones). Methanol extracts often have higher antioxidant activity. However, ethanol's performance is also effective, usually being close to that of methanol, albeit usually lower in terms of phenolic and flavonoid yield. (34)

After grinding the plants and adding methanol, the sample contained in the Erlenmeyer flask was subjected to ultrasound equipment to homogenize the sample more efficiently. Finally, the extract was subjected to thin-layer chromatography (TLC) analysis to verify the plurality of compounds present in the plant extract sample.

The TLC analysis used a silica gel sample as the stationary phase, which is a porous material with high capillarity, capable of attracting the mobile phase (35). Thus, the stationary phase (SiO_2) can attract the compounds present in the extract, according to the solvent used, which was methanol, mixed in a ratio of 1:2 with ethyl acetate. Each silica plate was placed in a dark chamber with a power of 110V, containing lamps that emit radiation at 2 different wavelengths, 254nm and 365 nm.

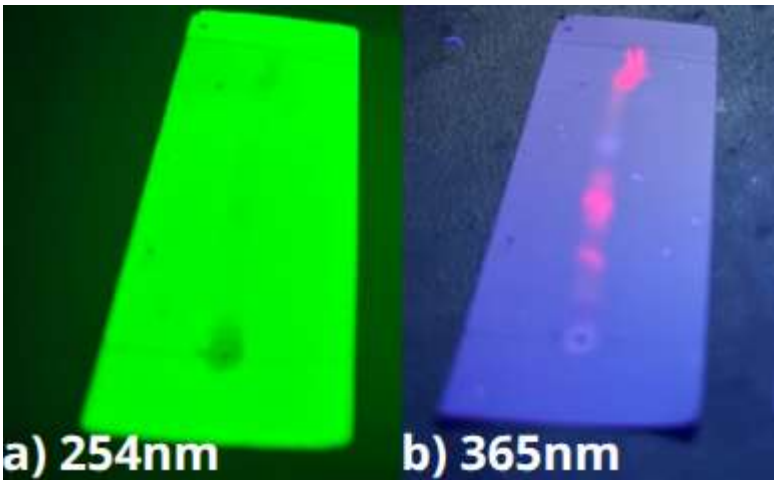


Figure 12 - TLC analysis of Neem on methanol a) 254nm; b) 365nm.

Figure 12 shows the results of the thin layer chromatography analysis of the extracts using methanol solvent placed in a dark chamber with different wavelengths. As indicated by the plates photographed at 254 nm (left) and 365 nm (right), neem extracts included the presence of flavonoids, evidenced by a pink glow under UV light. According to Souza (2023), compounds with low molecular weight, such as terpenes and flavonoids, are found in higher concentrations and may be related to the CCD profile found. According to Vani (2016), neem is a plant with several constituent varieties, among which the measured tetranortriterpenoids, including nimbin, nimbinin, nimbidinin, nimbolide and nimbidic acid, ascorbic acid, amino acids, and azadirachtin. It is noteworthy, however, that the flavonoid group represents, within the described proportion, a more significant portion when compared to other listed organic compounds. Flavonoids can act as antioxidants, whose

free radicals decrease interaction with oxygen, and this factor may have influenced the results.

6.4 Data Collection

The seeds were planted on March 28, 2025, and data collection was conducted every week, for 10 weeks, starting on April 4. Thus, data collection happened on the following dates: April 4, April 12, April 19, April 26, May 3, May 10, May 17, May 24, May 31 and June 7. The experiment was run at the Biotechnology Laboratory of Colegio Dante Alighieri, in São Paulo, Brazil (exact location: 23° 33' 1.872'' S and 46° 37' 59.9124'' W).

Ten vases, numbered 1 to 10 in each group after the seeds started to germinate (a few seeds did not germinate, as expected, as warned by the supplier), were used to grow the plants. The plants were measured (stem length) with a standard ruler with units in centimeters at the end of each week. An important fact to be considered was that the school closes on Sundays, and the lab, being a secure facility inside the school, was not available, and the plants could not be watered on those days. Also, Good Friday (April 18) is a national holiday in Brazil, which also prevented watering on that day.

The summary of the data is displayed in Tables 04 and 05, which show the sizes reached (in cm) by the plants in both the Neem and Control groups during the data collection period. In addition, Figure 13 shows the plants after 10 weeks of growth.



Figure 13 - Results of a) Neem and b) Control Group over the 10 weeks

Table 04 - Size of plants in centimeters per bell pepper vase for the Control group (COG) each week.

Control Group (COG) size (cm)										
Vase	***Week 1***	***Week 2***	***Week 3***	***Week 4***	***Week 5***	***Week 6***	***Week 7***	***Week 8***	***Week 9***	***Week 10***
1	0.60	0.80	1.30	1.80	2.20	2.90	3.50	3.90	4.60	5.20
2	1.20	1.60	2.00	2.60	2.90	3.00	3.40	3.80	4.40	5.10
3	0.80	1.10	1.60	3.00	2.80	3.10	3.60	4.10	4.50	5.10
4	0.70	0.90	1.20	1.50	1.90	2.60	3.00	3.40	3.80	4.20
5	1.40	2.50	2.90	3.10	3.40	3.70	4.00	4.20	4.80	5.40
6	0.90	1.40	1.70	2.00	2.40	2.70	3.20	3.80	4.30	4.50
7	0.60	1.00	2.00	2.30	2.50	2.80	3.00	3.20	3.50	3.90
8	1.80	2.30	2.50	2.70	3.60	4.10	4.30	4.40	4.90	5.20
9	1.60	2.10	2.40	2.60	3.00	3.50	3.70	3.90	4.30	4.50
10	1.20	1.40	1.70	1.80	1.90	2.10	2.40	3.10	3.40	3.70

Source: Authorship

Table 05 - Size of plants in centimeters per bell pepper vase for the Control group (COG) each week.

Neem Group (NGR) size (cm)										
Vase	***Week 1***	***Week 2***	***Week 3***	***Week 4***	***Week 5***	***Week 6***	***Week 7***	***Week 8***	***Week 9***	***Week 10***
1	1.50	1.70	2.50	3.00	4.70	5.70	5.90	6.00	6.20	6.40
2	1.90	2.10	2.70	3.40	4.00	4.80	5.00	5.40	5.50	5.60
3	0.60	0.90	1.20	2.40	2.80	3.40	4.30	5.20	5.30	5.50
4	0.80	1.30	1.30	1.50	1.70	1.90	2.40	2.90	3.60	4.10
5	1.10	1.70	2.40	2.70	3.00	3.60	4.40	4.80	5.10	5.40
6	1.60	2.40	3.20	4.00	4.70	5.30	5.60	5.70	6.00	6.20
7	0.80	1.50	2.30	2.70	3.00	3.40	3.80	4.40	4.90	5.30
8	1.10	1.30	2.10	2.40	2.90	3.00	3.50	3.60	3.90	4.60
9	1.90	2.40	2.90	3.70	4.30	4.70	5.00	5.80	6.30	6.60
10	2.00	2.60	3.20	3.50	3.70	4.10	4.40	4.50	4.90	5.10

Source: Authorship

Figure 13 – Photograph showing the aspect of plants in both groups after 10 weeks of growth.

6.5- Statistical Analysis

The data presented in tables 04 and 05 allowed a 2-way ANOVA analysis with repeated measures to be performed. Initially, the mean, variance, and standard deviation were obtained, as well as other relevant information, as shown in Table 06. The mean for each group during the experimental period is also shown in the bar graph in Figure 14.

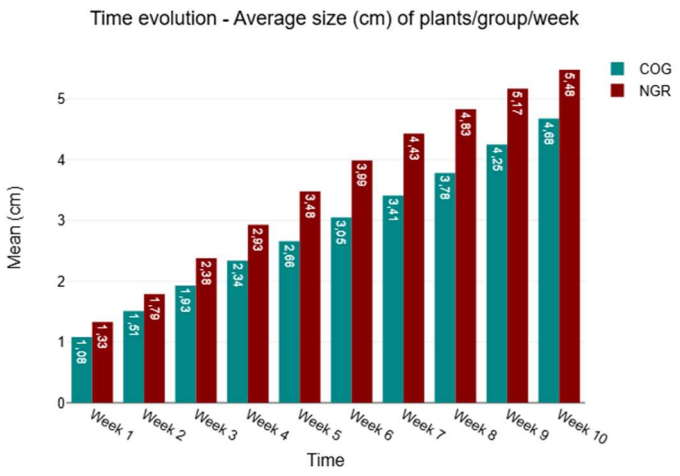


Figure 14 - Bar graph showing time evolution of plant size (cm) per group

Table 06 - Descriptive statistics of weekly plant height in COG and NGR groups

Week	Group	Frequency	Mean	Std. Deviation	Variance	Minimum	Maximum	Range	95% Confidence interval for mean
Week 1	COG	10	1.08	0.43	0.18	0.6	1.8	1.2	0.78 - 1.38
	NGR	10	1.33	0.52	0.27	0.6	2	1.4	0.96 - 1.7
Week 2	COG	10	1.51	0.6	0.37	0.8	2.5	1.7	1.08 - 1.94
	NGR	10	1.79	0.56	0.32	0.9	2.6	1.7	1.39 - 2.19
Week 3	COG	10	1.93	0.54	0.29	1.2	2.9	1.7	1.54 - 2.32
	NGR	10	2.38	0.7	0.49	1.2	3.2	2	1.88 - 2.88
Week 4	COG	10	2.34	0.55	0.3	1.5	3.1	1.6	1.95 - 2.73
	NGR	10	2.93	0.75	0.56	1.5	4	2.5	2.4 - 3.46
Week 5	COG	10	2.66	0.59	0.34	1.9	3.6	1.7	2.24 - 3.08
	NGR	10	3.48	0.97	0.93	1.7	4.7	3	2.79 - 4.17
Week 6	COG	10	3.05	0.58	0.34	2.1	4.1	2	2.63 - 3.47

	NGR	10	3.99	1.16	1.33	1.9	5.7	3.8	3.16 - 4.82
Week 7	COG	10	3.41	0.54	0.3	2.4	4.3	1.9	3.02 - 3.8
	NGR	10	4.43	1.03	1.06	2.4	5.9	3.5	3.69 - 5.17
Week 8	COG	10	3.78	0.43	0.18	3.1	4.4	1.3	3.48 - 4.08
	NGR	10	4.83	1	1.01	2.9	6	3.1	4.11 - 5.55
Week 9	COG	10	4.25	0.52	0.27	3.4	4.9	1.5	3.88 - 4.62
	NGR	10	5.17	0.91	0.82	3.6	6.3	2.7	4.52 - 5.82
Week 10	COG	10	4.68	0.6	0.36	3.7	5.4	1.7	4.25 - 5.11
	NGR	10	5.48	0.78	0.61	4.1	6.6	2.5	4.92 - 6.04

The statistical analysis has shown that plants treated with Neem extract presented a mean stem length significantly longer than those in the Control group during the whole experimental period. A mixed model ANOVA incorporating both groups and analyzing repeated measures on the same plants was performed to test whether there was a significant difference between the groups along 10 weeks in relation to the dependent variable, a significant difference between the groups in relation to the dependent variable, and whether there was an interaction between the two factors (time and treatment group) in relation to the dependent variable. The results of the ANOVA are shown in Table 07.

Table 07 - ANOVA

	Sum of squares	df	Mean Square	F	p	η^2	η_p^2
Week 1, Week 2, Week 3, Week 4, Week 5, Week 6, Week 7, Week 8, Week 9, Week 10	309.21	9	34.36	259.09	<.001	0.72	0.94
Group	25.35	1	25.35	6.38	0.021	0.06	0.26
RM Factor x Group	4.04	9	0.45	3.38	0.001	0.01	0.16
Residuals (Between Subjects)	71.47	18	3.97				
Residuals (Within Subjects)	21.48	162	0.13				

The analysis has shown that plant sizes have grown significantly over the 10-week period. Thus, the first null hypothesis, that there is no difference across these weeks, was rejected with a p-value <0.001.

The very small p-value raises a question. Is this value a reflection only of the natural phenomenon of plant growth (if water, sunlight and nutrients are available) – or does it also reflect a contribution of the experimental setup, beyond the natural baseline? The p-value does not distinguish between these factors, and the ANOVA merely tests whether the repeated measurements show variability in size, without isolating natural growth from other factors. In fact, a "no-growth" control group would be necessary to isolate the factors, which is unfeasible. The experimental setup likely provided favorable conditions for plant growth and for the performance of a controlled experiment (same light and water schedule). Thus, what can be said is that the controlled environment was important for the experiment's success, allowing for the comparison between the experimental groups.

The second null hypothesis analyzed was that there is no significant difference between the Neem and control groups in relation to plant size. This hypothesis was also rejected, with a p-value of 0.021. There was a noticeable difference in the average size of plants between the group grown with water only and the group treated with Neem extract when looking at all weeks taken together. The Neem plants were generally larger.

The third null hypothesis was that there was no interaction effect between the time factor (Weeks 1 to 10) and the two experimental groups. This hypothesis was also rejected, since the way the plants grew over time was different between the two experimental groups. The p-value for "repeated measures versus group" interaction is 0.001. Not only did the Neem group start bigger and stayed that way, but also it grew faster compared to the Control group.

The average growth rate for the control group was ~0.390 cm per week, whereas the Neem treatment added ~0.0839 cm per week compared to the Control group, resulting in an average growth rate of 0.474 cm/week. Figure 15 shows each plant's individual growth, as well as the average growth rate for each experimental group and the spread of the data, in orange for the Control group and blue for the Neem-treated group.

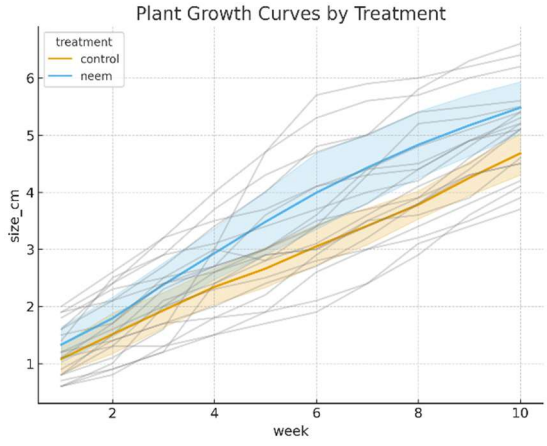


Figure 15 - Plant Growth Curves by Treatment.

7. CONCLUSION AND DISCUSSION

This study has sought to test the effectiveness of a candidate “nature-based” fertilizer, the Indian Neem. As discussed above, this plant contains compounds with fertilizer and pesticide properties and has the potential to supplement the usage of NPK fertilizers, which can pollute the water supply due to misuse. The present data investigated only the potential effects of Neem as a fertilizer, not as a pesticide, which would involve the use of agricultural pests and would complicate the experimental design.

As demonstrated in the Statistical Analysis session, the ANOVA suggests that treatment with the Neem extract may influence plant size beyond just providing water, as happened in the control group. The data shows that by week 10, the average size in the Neem group was ~5.48 cm compared to ~4.68 cm in the Control group. This overall difference suggests that Neem treatment had a beneficial impact on plant growth. It could mean that the extract helped the plants start bigger, grow more consistently, or both. However, the analysis does not specify when or how the difference developed. Finally, the partial eta-squared ($\eta p^2 = 0.26$) indicates that approximately 26% of the variability in plant size between the Neem group and the Control group can be explained by whether the plants received Neem or not. Thus, the effect was moderate, indicating that the difference is likely not caused only by the addition of the Neem extract and other factors might have contributed to the observed result.

The analysis also indicated that the pattern of growth over the 10 weeks was not the same for the Control and Neem groups. In other words, the Neem plants did not just start larger and stay that larger: they grew differently or faster over time compared to the Control plants. Thus, the effect of Neem may build or change as the experiment progresses. This finding suggests that the Neem treatment does not provide a one-time boost but actively changes how the plants develop over time. As shown by the data, the gap in size between the groups widened from ~0.25 in week 1 to 0.80 cm by week 10, indicating that Neem might accelerate growth or sustain it better. This interaction suggests that Neem has a dynamic effect, not just a static one. The partial eta-squared ($\eta p^2 = 0.16$) means that approximately 16% of the variability in growth patterns can be attributed to the interaction between Neem treatment and time. This is also a moderate effect but reinforces that the difference in growth trajectories is noticeable (Figure 15).

In addition, the TLC technique performed in a partner laboratory showed that Indian Neem leaves have a wide range of compounds, observed by the distinct variation in colors in the dark chamber, i.e., they have molecules with different absorption energies, which alters the visible color of a complex plant, since each spot color is a compound. The pink spots (which are in the majority) showed signs of being flavonoids, which absorb photons around 350 to 400 nm, which may explain their significant fluorescence at 365 nm. Flavonoids act as pigments, protect against UV rays, insects, fungi, and other harmful agents, and can neutralize free radicals, which can help prevent various diseases.

As discussed in the Introduction, the role of artificial fertilizers in increasing the productivity of agro-industry crops is well established but presents environmental problems. Therefore, the Indian Neem was selected to test and/or develop a new product, "AGRO-BHARAT", with the potential to provide the necessary nutrients for the growth and development of bell pepper plants (selected because of their widespread use and their sensitivity to artificial fertilizers and pesticides), enhancing agricultural productivity.

Taken together, the present results indicate that the Neem extract was better than just watering the plants and changed how the plants grew over time, warranting the development of further studies to investigate Neem as a candidate for boosting plant development.

Though promising, the scalability of neem-based products as fertilizers for global vegetable production is constrained by current cultivation, logistics, and competing demands (37). According to FAO, global vegetable harvested area reached approximately 131 million acres in 2024, underscoring the immense scale needed for widespread adoption to supplant synthetics. Neem is currently cultivated across an estimated 1.73 million acres worldwide, with India accounting for ~1.48 million acres. Mature neem plantations yield an average of 0.8 tons of seeds per acre annually, translating to a theoretical global production of 1.4 million metric tons of seeds per year. For vegetable applications, a conservative estimate requires approximately 10 kg/acre on average, yielding a total demand of 1.3 million tons if Neem were to fully supplant conventional inputs across all vegetable acreage. However, agriculture currently consumes only 50–60% of neem output, with the remainder allocated to cosmetics and pharmaceuticals/food additives. This leaves an effective agricultural supply of approximately 770,000 tons—covering just 59% of the hypothetical full-scale vegetable demand and highlighting a potential shortfall of 530,000 tons that could intensify due to competing demand. In the short term, full global coverage remains unrealistic, necessitating targeted implementation, such as prioritizing high-value organic vegetables or hybrid systems blending neem with synthetics (38). Long-term viability supports expansion: Neem's low-input, drought-tolerant nature allows rapid scaling (maturity in 5–7 years), with

Indian initiatives like the National Medicinal Plants Board already adding 300,000 acres annually, potentially doubling global area to 3.5 million acres and yielding 2.8 million tons to meet competing demands. Enhanced yields via improved varieties, as well as local cultivation in places such as subtropical South America (where neem adapts well) could increase output by 20–30% (39) (40). Ultimately, while current supplies fall short for universal vegetable application, strategic investments in cultivation and processing could render neem economically viable within 5–10 years, supporting global sustainability goals.

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